

MOHIT SONI

B.Tech. Computer Science and Engineering

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LinkedIn

GitHub

EDUCATION

Indian Institute of Information Technology, Guwahati
B.Tech. in Computer Science and Engineering **CGPA: 8.66/10.0**

2024 – 2028

Board of Secondary Education, Rajasthan
Senior Secondary (Class XII) **88.4%**

India

PROJECTS

Zarithm – Sales & Inventory Management Backend

May 2026

Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt

GitHub

- Engineered a RESTful backend in Node.js/Express with clean MVC separation, exposing 15+ API endpoints across 6 resource modules (authentication, products, sales, customers, stock logs, dashboard) for a sales and inventory management system targeted at small businesses.
- Modeled schemas in MongoDB via Mongoose to support customer credit/dues tracking, stock-level logging, and aggregation-driven dashboard analytics.
- Implemented JWT-based authentication with bcrypt password hashing, cookie-based sessions, and CORS-restricted cross-origin access control.
- Automated user data lifecycle management with a scheduled background job handling pending account deletion and cleanup.

Lunar Lander Autopilot – Genetic Algorithm RL Agent

2026

Python, NumPy, OpenAI Gymnasium

GitHub

- Designed a from-scratch Genetic Algorithm, in place of gradient-based reinforcement learning, to train an 8-16-4 feed-forward neural network policy (288 parameters) for OpenAI Gymnasium's LunarLander-v3 environment.
- Evolved a population of 100 policies over 500 generations using elitism (top 20 retained), single-point crossover, and adaptive Gaussian mutation with a decaying standard deviation to balance exploration and convergence.
- Formulated a variance-aware fitness function (mean reward minus a penalty on reward variance across 20 episodes) to select for consistent, low-risk landings over high-variance outliers.
- Achieved a best evaluation reward of 297, validated through deterministic (greedy) inference over 100 held-out episodes.

TAP-TAP – Typing Speed & Accuracy Platform

August 2025

HTML, CSS, JavaScript (Vanilla)

GitHub

- Built a Monkeytype-inspired typing test in vanilla JavaScript with no external frameworks, supporting configurable time-based and word-count-based practice modes.
- Implemented real-time, character-level error highlighting by diffing expected versus typed input on every keystroke event.
- Engineered a live scoring engine computing Words Per Minute (WPM), accuracy percentage, and raw typing speed at session completion.

SCHOLASTIC ACHIEVEMENTS

- Competitive Programming:** Solved 510+ problems on LeetCode (**rating: 1593**) and 550+ problems cumulatively across LeetCode, GeeksforGeeks, and Codeforces.
- Smart India Hackathon (SIH) 2025 – Ranked Top 7 Nationally:** Placed 2nd on the national waitlist, one position behind qualification for the on-site Grand Finale (Top 5 teams advanced).

TECHNICAL SKILLS

- Languages:** C, C++, Java, JavaScript, Python
- Backend & APIs:** Node.js, Express.js, RESTful API Design, JWT Authentication, MVC Architecture, Middleware Design
- Databases:** MySQL, MongoDB, Mongoose ODM, Schema Design & Indexing
- Frontend:** HTML5, CSS3
- Machine Learning & Data Science:** NumPy, Pandas, Matplotlib
- Tools & Platforms:** Git, GitHub, Postman, VS Code, Figma, Linux/Unix
- Also Familiar With:** React.js, Tailwind CSS, Prolog

COURSEWORK

Computer Science Coursework: Database Management Systems, Data Structures & Algorithms, Operating Systems, Object-Oriented Programming (OOP), Artificial Intelligence